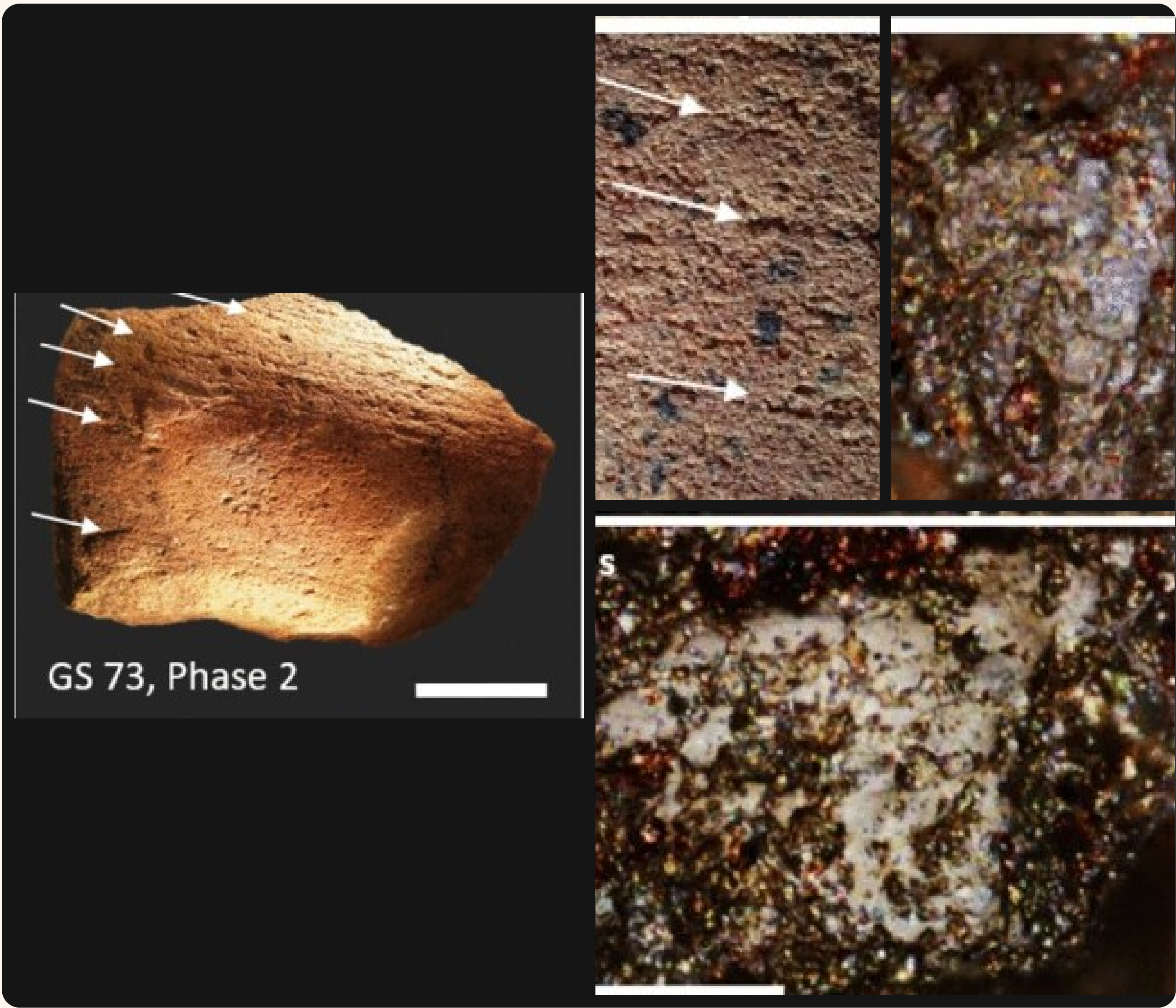


What can one stone tell us?

Look closely first. The source record gives observations and measurements, not the answer.



What does GS73 suggest about life at Madjedbebe in ancient Australia?

THE ANCIENT OBJECT

GS73, a sandstone millstone fragment excavated from square B5/52 at Madjedbebe on Mirarr Country.

MODERN RECORD & AUDIENCE

Hayes and colleagues published photographs, micrographs and laboratory results in a peer-reviewed 2022 study to document the object and argue for long-term grinding-stone use.

EXCAVATION CONTEXT

Phase 2, modelled to 68.7-50.4 thousand years ago. The earliest end of this range is debated.

WEAR & RESIDUE EVIDENCE

Deep grooves, levelled quartz grains, striations and bright polish; 41 starch grains and plant-related compounds. Exact plant unidentified; plastic contamination detected.

COMPARISON RECORDS

Clarkson et al. defend the early depositional context. Williams et al. use termite and sediment evidence to question whether the earliest material stayed in place.

Five questions historians ask

Each question adds a different kind of thinking to the answer.

HISTORICAL QUESTION	THE REASONING JOB	GS73 EXAMPLE	SENTENCE PATTERN
1 What is it?	Identify type, origin and purpose. Separate the ancient object from the modern record.	GS73 is the ancient object. The 2022 figure is a modern archaeological record.	GS73 is... The figure was created to...
2 What can I see?	Record visible or measured details before explaining them.	Deep grooves, levelled grains, striations, bright polish and 41 recovered starch grains.	The source shows... The study recorded...
3 What was happening?	Use accurate knowledge about time, place and practice to make the details meaningful.	Grinding made hard or fibrous resources easier to prepare and use.	In this context... This mattered because...
4 What does it suggest?	Answer the inquiry and connect each inference to evidence.	Grooves, wear and plant-related results support repeated plant processing.	The evidence suggests... because...
5 How sure am I?	Judge usefulness, reliability, uncertainty and disagreement.	Several methods agree about function; the exact plant, user and earliest date remain uncertain.	This is useful because... However...

Origin, features & purpose
What is it and why was it recorded?

Historical context
What was happening?

Interpretations + evidence
Who argues what, using which details?

Usefulness & reliability
How far can we trust the claim?

Metacognition
What will I check and revise?

Worked examples above. Learning continuum below.

Read across to see 12-month growth. Read down to connect each coloured phrase to its matching continuum cell.

YEAR LEVEL	YEAR 5	YEAR 6	YEAR 7	YEAR 8	YEAR 9	YEAR 10
WORKED EXAMPLE	GS73 is a stone source found at Madjedbebe. I can see deep grooves and shiny patches. It comes from ancient Australia and may have been used for grinding. Researchers disagree about exactly how old its soil layer is. The stone is useful for showing tool use, but it cannot tell us the user's name. I checked my answer and added the words 'may have' because the source does not prove everything.	GS73 is an archaeological artefact: a sandstone grinding-stone fragment found at Madjedbebe on Mirarr Country. Photographs show grooves and polish. It belongs to a very early period of Aboriginal occupation, when people repeatedly used stone tools. Some researchers place Phase 2 at about 65,000 years ago, while others question whether soil movement affected that date. It is useful evidence that people ground materials, although the stone alone cannot identify exactly what was ground. Before writing, I broke the task into source, context, interpretations and usefulness.	The ancient source is GS73, a sandstone millstone fragment excavated from Phase 2 at Madjedbebe on Mirarr Country. The modern record is a 2022 scientific figure by Hayes and colleagues, created to document its wear and residues and argue for long-term grinding-stone use. In ancient Australia, grinding technology helped people prepare hard or fibrous plant foods and shows detailed knowledge of local resources. Hayes and colleagues interpret plant processing because the source contains deep grooves, microscopic polish, 41 starch grains and plant-related compounds; other researchers question the earliest date of the layer. GS73 is highly useful for investigating plant processing because shape, wear and residues agree, but the exact plant is unidentified and the precise age remains contested. After reviewing, I changed 'proves' to 'supports' and separated the well-supported function from the disputed date.	The ancient source is GS73, a millstone fragment from Phase 2; Hayes and colleagues' 2022 scientific figure records its form, wear and residues so readers can examine their interpretation. Madjedbebe is a long-occupied site on Mirarr Country, and the grinding evidence places complex plant preparation within deep Aboriginal history. Hayes interpret seed grinding from wear and residues; Clarkson defend the early context using site integrity and dating; and Williams use termite and sediment evidence to make the precise age less secure. Shape, microscopic wear, starch and plant compounds strengthen the function claim, but I would ask: Could contamination affect the residues? Could layer movement affect the date? I reflected by checking which methods added independent support and recording the questions still unresolved.	GS73 is material evidence excavated from a recorded Phase 2 context. The peer-reviewed 2022 figure combines photographs, micrographs and laboratory results to persuade a scientific audience that grinding-stone use has a deep history. The study belongs to research into the deep Aboriginal history of Sahul and was conducted with permission from Mirarr custodians. Hayes use grooves, polish and residues to argue for seed processing; Clarkson use site integrity and dating to support the early chronology; while Williams use termite and sediment evidence to challenge it. Agreement between wear and residue evidence makes grinding the strongest claim, while the exact plant and precise age are weaker because they depend on preservation and depositional context. I monitored whether I was comparing evidence or merely listing it, then rewrote the list as a judgement.	GS73 is an ancient artefact interpreted through a modern scientific publication. Its explicit evidence is recorded wear and residues; its implicit argument is that grinding technology has a deeper Australian history than older accounts allowed, aimed at archaeologists and other scientific readers. The investigation reflects contemporary scientific testing and collaboration with Mirarr custodians, revealing values of empirical testing, custodial permission and careful qualification. Hayes and Clarkson accept the early context because several records converge, while Williams gives more weight to termite disturbance and sediment movement; their interpretations differ because they make competing judgements about site formation. Overall, GS73 is strong evidence for grinding but only qualified evidence for an exact age: strengths include converging methods; limitations include unidentified starch, contamination and contested layer history. I justified triangulation as the strongest strategy, then adapted the conclusion because the claims have different certainty.
WHY IT MEETS THIS YEAR	■ Origin, features & purpose: The blue phrases name the source and two visible features: 'stone source', 'deep grooves' and 'shiny patches'. Origin and purpose are not yet developed. ■ Historical context: The green phrase places GS73 broadly in 'ancient Australia' but gives little detail about life or practice. ■ Historical interpretations: The orange phrase notices disagreement about the date without describing the competing arguments. ■ Accuracy, usefulness & reliability: The purple phrases give one use and one limit: tool use is visible, but the user cannot be named. ■ Metacognition: The grey phrase shows a real revision: adding 'may have' makes the inference appropriately cautious.	■ Origin, features & purpose: 'Archaeological artefact', 'sandstone grinding-stone fragment' and 'grooves and polish' identify type and features; purpose is not yet explained. ■ Historical context: 'A very early period of Aboriginal occupation' identifies the context and connects it to repeated tool use. ■ Historical interpretations: The answer recognises two positions on the date but does not yet describe the evidence behind them. ■ Accuracy, usefulness & reliability: It recognises usefulness for grinding and a clear limit: the material processed cannot yet be identified. ■ Metacognition: The response was planned by breaking the task into the four historical-thinking parts shown on the wall.	■ Origin, features & purpose: The answer separates the ancient object from the modern 2022 record, then identifies origin, features and why the figure was created. ■ Historical context: The green sentence outlines how grinding supported plant preparation and knowledge of local resources. ■ Historical interpretations: The orange phrases describe Hayes' plant-processing interpretation and the separate challenge to the earliest date. ■ Accuracy, usefulness & reliability: The purple judgement concludes that converging methods make plant processing useful evidence, then bounds the claim with two limits. ■ Metacognition: The grey phrase makes editing visible: 'proves' becomes 'supports', and function is separated from precise age.	■ Origin, features & purpose: The blue phrases describe the object and explain how the modern figure lets readers inspect its form, wear and residues. ■ Historical context: The green phrases describe Madjedbebe, Mirarr Country and grinding within deep Aboriginal history. ■ Historical interpretations: The answer explains the debate by connecting each interpretation to a different evidence base. ■ Accuracy, usefulness & reliability: It compares independent evidence for function and asks targeted reliability questions about contamination and layer movement. ■ Metacognition: The reflection identifies which methods added independent support and preserves unresolved questions for further inquiry.	■ Origin, features & purpose: The blue phrases explain why the peer-reviewed figure combines several records and which audience and argument it addresses. ■ Historical context: The green phrases explain the deep-history research context and the role of Mirarr custodial permission. ■ Historical interpretations: Each interpretation is analysed through its evidence: wear and residue, site integrity and dating, or termite and sediment movement. ■ Accuracy, usefulness & reliability: The purple comparison identifies grinding as a stronger claim than exact plant or exact age and explains why. ■ Metacognition: The writer monitors whether evidence is merely listed, then deliberately revises the list into a comparative judgement.	■ Origin, features & purpose: The blue phrases analyse explicit evidence, the publication's implicit argument, purpose and scientific audience. ■ Historical context: The green phrases use context to explain the creators' values: empirical testing, permission and qualification. ■ Historical interpretations: The evaluation exposes why interpretations differ: researchers give different weight to converging records and site-formation evidence. ■ Accuracy, usefulness & reliability: The overall judgement weighs a strength, three limitations and two claims with different degrees of certainty. ■ Metacognition: The writer justifies triangulation, then adapts the conclusion rather than forcing all claims to sound equally certain.
SKILL AREA	YEAR 5	YEAR 6	YEAR 7	YEAR 8	YEAR 9	YEAR 10
Origin, features & purpose	I can name a historical source and a feature I notice.	I can identify different types of historical sources.	I can identify the origin, content features and purpose of historical sources.	I can describe the origin, content features and purpose of historical sources.	I can explain the origin, content features and purpose of historical sources.	I can analyse the origin, explicit and implicit meaning, and purpose of historical sources.
Historical context	I can say where or when a source comes from.	I can identify the historical context of a source.	I can outline the historical context of a source.	I can describe the historical context of sources.	I can explain the historical context of a source, including important events that influenced its creation.	I can explain historical context to identify the motivations, values and attitudes of a source's creator.
Historical interpretations	I can notice that people may explain the past differently.	I can recognise that historians may interpret historical events and developments differently.	I can describe different historical interpretations of the past.	I can explain different historical interpretations and contested debates about the past.	I can analyse historical interpretations and contested debates and identify the evidence used to support them.	I can evaluate historical interpretations and contested debates, including why or how interpretations differ.
Accuracy, usefulness & reliability	I can say one thing a source can and cannot tell us.	I can recognise that some historical sources are more useful than others.	I can draw conclusions about the usefulness of sources.	I can compare and contrast historical sources and ask questions about their accuracy, usefulness and reliability.	I can compare and contrast historical sources and identify their accuracy, usefulness and reliability.	I can compare and contrast sources and analyse their accuracy, usefulness and reliability.
Metacognition	I can follow a source-analysis checklist one step at a time.	I can complete the task by breaking it into smaller parts.	I can review and edit my work.	I can reflect on my learning experience and processes to inform future progress.	I can use, monitor and evaluate a variety of learning strategies.	I can justify and adapt the source-analysis strategies that produced my strongest judgement.

Strong evidence can still support a contested claim

Historians and archaeologists may agree about what a tool did while disagreeing about exactly when it was used.

WHAT WAS GS73 USED FOR?

Hayes and colleagues, 2022

The team interprets GS73 as a seed-grinding tool. The conclusion draws on deep grooves, microscopic usewear, 41 recovered starch grains and plant-related chemical compounds.

What must stay qualified:
Several independent methods point towards plant processing. The exact plant was not identified, and plastic contamination was also detected.

HOW OLD IS PHASE 2?

Two archaeological interpretations

Clarkson and colleagues argue that artefacts remained in primary context and support occupation about 65,000 years ago. Williams, Spooner, McDonnell and O'Connell argue that termite activity and sediment disturbance may have moved artefacts and make the earliest date less secure.

What must stay qualified:
The dispute concerns the integrity and dating of the layer, not whether GS73 is a real grinding stone.

Revision changes the claim:
Too certain: GS73 proves people ground seeds 65,000 years ago.
Better: Several lines of evidence strongly support seed grinding, while the precise age of the Phase 2 context remains contested.